24PIT302 **JAVA PROGRAMMING**

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PREREQUISITES: C Programming

COURSE OBJECTIVES:

- To Learn the fundamental concepts of Java programming
- To Acquire the knowledge of inheritance, abstraction, exception and package in Java
- To Obtain the knowledge of Java Collection API, Multithreading, JDBC and Lambda expression in Java

UNIT I INTRODUCTION TO JAVA **9**

History of Java – Features of Java – Java Architecture – Comments – Data Types – Variables – Operators – Type Conversion and Casting – Flow Control Statements – Reading Input from keyboard – Command Line Arguments – Using Scanner Class – Arrays – Classes and Objects – UML Class diagram – Methods – Constructors – static variables and Methods – this Keyword – Encapsulation – Concept of Access Control.

UNIT II INHERITANCE **9**

Inheritance – Types of Inheritance – Super and Sub Classes – super keyword – final class and methods – Object class –Polymorphism – Types of polymorphism – Method Overloading – Constructor Overloading – Method Overriding – Dynamic Method Dispatching – garbage collection – String class –String Buffer class –String Builder class.

UNIT III DATA ABSTRACTION **9**

Packages – Introduction to Packages – User Defined Packages – Accessing Packages – Abstract classes and Methods – Interface – Defining an interface – implementing interfaces – extending interfaces – Multiple Inheritance Using Interface – Exception Handling – Errors vs Exceptions – Exception hierarchy – usage of try – catch – throw – throws and finally – built in exceptions – user defined exceptions.

UNIT IV COLLECTION API AND LAMBDA **9**

Introduction to wrapper classes – Predefined wrapper classes – Conversion of types – Concept of Auto boxing and unboxing – Java Collections API – Introduction to Collection – Generics – List implementations – Set implementations – Map implementations – Functional Interfaces – Lambda Expressions – Accessing local variables – Accessing class variables – Predicates – Functions – Suppliers – Consumers – Stream API – Filter – Sorted – Map – Reduce – Count – Parallel Streams.

UNIT V JDBC AND MULTITHREADING **9**

JDBC – Introduction to JDBC – Establishing connection – Executing query – Processing results – Prepared Statement – Callable Statement – Transactions – Meta Data objects – Multithreading:

Introduction to Multithreading – Process Vs Thread –Thread life cycle – Thread class – Runnable Interface – Thread creation – Thread control and priorities – Thread synchronization.

THEORY: 45PERIODS

LIST OF EXPERIMENTS:

1. Develop programs using flow control statements and arrays to manage execution flow and data organization effectively.
2. Implement programs using inheritance and polymorphism to promote code reusability and dynamic method binding.
3. Develop programs incorporating packages, abstract classes, and interfaces to structure code modularly and enforce abstraction.
4. Implement programs using exception handling mechanisms to ensure robust error detection and graceful recovery.
5. Create programs using the Collection API to manage groups of objects with flexibility and high performance.
6. Implement programs using JDBC to establish and manage database connections for data persistence and retrieval.
7. Develop programs using multithreading to achieve concurrent execution and improve application performance.
8. Case Study.

LABORATORY:30 PERIODS
TOTAL : 45+30= 75 PERIODS

COURSE OUTCOMES:

At the end of the course the students will be able to

CO1:	Infer the fundamental concepts, architecture, and features of Java Programming	**K2**
CO2:	Solve programming challenges using object-oriented paradigms	**K3**
CO3:	Build applications using multi-tasking mechanisms, and exception handling strategies	**K3**
CO4:	Construct robust and efficient Java applications using JDBC, lambda expressions and interface	**K3**
CO5:	Develop Java applications by amalgamating object-oriented design, collection usage and advanced data manipulation	**K3**

TEXT BOOKS:

1.Herbert Scheldt, " Java: The Complete Reference”, 12th edition, Tata McGraw-Hill, 2022.
2.Cay S Horstmann and Gary Cornell, "Core Java: Volume I – Fundamentals", 12th edition, Prentice Hall, 2021.

REFERENCES:

1.David Flanagan and Benjamin Evans, "Java in Nutshell", 8th edition, O’Reilly Media, 2022.

2.Kathy Sierra, Bert Bates, Trisha Gee, "Head First Java ", 3rd edition, O'Reilly Media, Inc, 2022.
3.Joshua Bloch, "Effective Java", 3rd Edition, Addison-Wesley Professional, 2018.

WEBURLs:

1.www.https://docs.oracle.com/javase/tutorial/java/nutsandbolts
2.www. https://javabeginner.com/learn-java
3.www. https://dev.java/learn

COURSE ARTICULATION MATRIX

CO'S	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	1	-	2	-	-	1	1	1	-	2	2	2	2
CO2	3	3	2	-	2	-	-	1	1	1	-	2	2	2	2
CO3	3	3	2	-	2	-	-	1	1	1	-	2	2	2	2
CO4	3	3	2	-	2	-	-	1	1	1	-	2	2	2	2
CO5	3	3	2	-	2	-	-	1	1	1	-	2	2	2	2
CO	**2.8**	**2.8**	**1.8**	**-**	**2**	**-**	**-**	**1**	**1**	**1**	**-**	**2**	**2**	**2**	**2**